



Prefabricated Building Modules for AI Data Centres: Technological Feasibility, Global Practice, and the Two-Level Stacked Data Hall

What MDC means, why the prefabricated building module is the family that delivers hyperscale AI capacity, how it differs from steel-frame construction and from container-scale MDC, why AI data halls can be stacked two storeys high, and why Grid Apex has adopted this method for its multi-storey AI data centre campus — illustrated from live prefabricated modular data centre programmes and referenced to projects across nine markets.

Jointly developed by **Grid Apex Inc.** gridapex.ai and **PSG Group** psggroup.net

The modular data centre has moved, in roughly five years, from an edge-computing curiosity to the default delivery method for multi-hundred-megawatt AI campuses. The market reflects it: modular data centre revenues stood at **US\$31.24 billion in 2025**, are put at US\$35.44 billion in 2026, and are forecast to reach **US\$116.53 billion by 2034** — a compound annual growth rate of **16.0% over 2026–2034**, or 15.8% measured from the 2025 base. North America alone accounted for 37.04% — some US\$11.57 billion — of the 2025 total.¹

That growth is not driven by cost. Factory integration and proprietary power and cooling assemblies carry a reported premium of 20–30% per megawatt against conventional site-built construction. It is driven by **time**, by **labour scarcity**, and increasingly by **thermal physics**. When an accelerator rack draws 100–200 kW and the roadmap points at 600 kW, the cooling system stops being a service routed through a building and becomes a structural element of it. Factories build that kind of assembly better than construction sites do.

This paper is about one specific answer to that problem: the **prefabricated building module** — the structural bay that *is* the building, manufactured complete with its frame, envelope, services and cooling, then set and connected on site. It is the family that delivers gigawatt-class AI capacity, it is the family that makes stacking an architectural option rather than a container-yard expedient, and it is the family that Japanese, Australian and Southeast Asian projects are converging on. The paper sets out what it is, how it differs from steel-frame construction and from container-scale MDC, what the delivery sequence actually looks like, who builds it, and — the question that decides land economics on a constrained site — whether AI data halls made this way can be stacked two levels high. **It is also the method Grid Apex has adopted for its own multi-storey AI data centre campus**, and section 11 sets out why that produces a permanent Japanese building capable of housing the current and next generations of AI hardware.

1 What MDC means — and which family matters

MDC — modular data centre — denotes a data centre whose functional building blocks are manufactured, assembled, integrated and *tested* in a factory, then transported to site and connected together. The defining characteristic is not that the building is made of repeated parts. It is that the **factory, not the site, is where integration happens**.

In a conventional project, a contractor erects a structure and then installs equipment inside it: switchgear arrives, is positioned, wired, commissioned; chillers are craned onto a roof and piped; containment is built around racks. Every one of those operations happens outdoors, in sequence, subject to weather, trade availability and site congestion. In a modular project,

those same operations happen on a production line under cover, in parallel with the site works, and the resulting unit is factory-acceptance-tested before it is ever loaded onto a truck.

Three families are commonly grouped under the term, and conflating them causes most of the confusion in the market.

THE THREE FAMILIES OF MODULAR DATA CENTRE			
TYPE	WHAT SHIPS	TYPICAL UNIT	WHERE IT DOMINATES
Container-scale (ISO)	A sealed, self-contained box with IT, power and cooling inside a transport-legal envelope	10 ft / 20 ft / 40 ft; 0.5–2 MW	Edge, rapid deployment, constrained or remote sites, phased capacity
Skid and room modules	Pre-integrated functional rooms — power skids, chiller plants, UPS and battery rooms, CDU plant — without the building	Plant room or “pod”; 1–5 MW	Large campuses where the building is conventional but the MEP is prefabricated
Prefabricated building modules	Structural bays that <i>are</i> the building — floor, frame, envelope and services as one manufactured element	Transport-sized structural bay, typically 3–4.5 m × 12–16 m; joined into halls of any size	Whole-campus delivery where speed, labour scarcity and density dominate

The third family is the subject of this paper. The distinction from the first is not one of size but of **structural role**. A container-scale module is a piece of equipment that happens to have walls; the building, if any, goes around it. A prefabricated building module *is* a piece of the building: its frame is primary structure, its floor is the data hall floor, its envelope is the external wall and fire-rated line, and its roof is either the roof of the finished hall or the deck that carries the level above. That is what allows prefabricated building modules to be assembled into continuous data halls of tens of thousands of square metres, and to be stacked — neither of which a yard of shipping containers can do.

A single project frequently uses all three families. NEXTDC's **S4 Sydney** — a 30,743 m² facility of 350 MW across two four-storey buildings, under construction by Multiplex for mid-2027 completion — is conventionally framed but uses **prefabricated mechanical plantrooms, prefabricated electrical switch rooms and prefabricated main services reticulation pathways**, alongside precast lift cores, columns and façade.² It is simultaneously the largest modular data centre project in Australia and not a “container” project at all.

2 Anatomy of a prefabricated building module

A prefabricated building module for an AI data hall is a steel-framed volumetric unit sized by what can move on the road — typically **3–4.5 m wide, 12–16 m long and 3.5–4.5 m high** — and designed to be joined to its neighbours so that many modules combine into one continuous hall. The unit that ships is therefore not the unit that operates: capacity is quoted per hall or per campus, not per module, and a data hall of 2,000 m² is simply a hall built from more of them. It is best understood as four systems manufactured as one object.

- **The chassis.** A welded steel frame — base grillage, corner and intermediate posts, roof frame — fabricated to millimetre tolerance. It is simultaneously primary building structure, the transport frame, the equipment support and the lifting rig, and it must satisfy all of those load cases independently.
- **The envelope.** Insulated composite panel or rainscreen on a fire-rated substrate, with the module-to-module joint detailed as a continuous fire, air and water line. Because the joint is the one element made on site, its design governs the quality of the finished building far more than the panel specification does.
- **The services layer.** Containment, busway or cabling, switchboards, cooling distribution — CDUs, manifolds, primary and secondary coolant headers — fire detection and suppression, monitoring and controls, all installed and terminated at fixed

interface points on the module face.

- **The interface.** Four families of connection cross the module boundary: **power, coolant supply and return, network, and controls**. These are the only genuinely site-made joints in the whole building, which is why they are engineered as repeatable quick-connect details rather than as drawn-and-measured site work.

THE STRUCTURAL INSIGHT

In conventional construction the frame is a container for equipment. In prefabricated modular construction the frame is **simultaneously structure, shipping crate, equipment chassis and lifting rig**, and it must satisfy four load cases: in-service gravity and seismic loads, transport dynamic loads, lifting and racking loads during craneage, and **stacking loads** where modules are set on one another. A module designed only for the in-service case will fail on the road; a module designed only for transport cannot be stacked. The stacking case is what separates a building module from a container.

Module dimensions must be resolved against **road and sea transport conditions together**. Japan's general vehicle limits under the Vehicle Restriction Order (車両制限令 Art. 3) are 2.5 m width, 3.8 m height, 12.0 m length, 20 t gross, 10 t per axle and 5 t per wheel, with a 12 m minimum turning radius. Two qualifications matter. Height rises to **4.1 m** on designated 高さ指定道路 and gross weight to **22–25 t** on expressways and 重さ指定道路 according to axle spacing; but the **12.0 m length is measured over the tractor and trailer together and has no designated-road exemption**. A 40 ft module is 12.19 m before the tractor is attached, so **module transport in Japan is a permit exercise from the first drawing**, under 特殊車両通行許可 or the confirmation scheme that replaced part of it in 2022, with permit ceilings for the standard exempted vehicle types running to 16.5 m and 36 t on expressways. Route and overhead clearances are verified case by case. Cross-border delivery additionally requires that **vessel type, loading and discharge method, lashing and salt-spray protection** be settled at the design stage, and that port transshipment, sailing schedules and the final road leg be planned as one sequence. Elsewhere in Asia and in Europe the transport envelope likewise has to be verified against the actual route rather than assumed. Where a manufacturer quotes a large single figure — Daiwa House's approximately 200 m² Module DPDC — that denotes a *design bay*, which must be divided into several shipping pieces and mated on site. Transport engineering is therefore part of the structural design, not a logistics afterthought: the specification should state explicitly the distinction between the design bay and the shipped piece, and carry the combined sea-and-road conditions into module dimensions, weight and **lifting-point design**.

3 How it differs from steel-frame construction

The distinction most often misunderstood is between a prefabricated modular data centre and a steel-frame building that happens to be erected quickly. Both use steel. The difference is what the steel is doing, and when the mechanical and electrical content arrives.

CONVENTIONAL STEEL FRAME AGAINST PREFABRICATED MODULAR DELIVERY		
DIMENSION	CONVENTIONAL STEEL FRAME	PREFABRICATED BUILDING MODULES
Sequence	Structure first, then fit-out. Trades follow one another on site.	Structure and fit-out happen together in the factory; site works run in parallel.
Where integration occurs	On site, outdoors, in sequence.	On a production line, indoors, repeatably.
What the steel does	Carries the building. Equipment is installed into the volume it encloses.	Carries the building <i>and</i> is the transport frame, the equipment chassis, the lifting interface and, where stacked, the load path for the level above.
Testing	Commissioned on site after installation. Defects found late, at maximum cost.	Factory acceptance tested before shipment. Defects found in the factory.
Site labour	High and sustained — a constraint shared by the Japanese, Korean, Australian, European and US markets.	Sharply reduced; site work is foundations, craneage, module-to-module joints and connection.
Tolerance regime	Construction tolerances, measured in centimetres.	Manufacturing tolerances, measured in millimetres — necessary for modules to mate.
Change flexibility	High until late. Design can evolve during construction.	Structure and principal interfaces frozen early; internal layout can still be assessed for adjustment.
Cost	Baseline.	Reported 20–30% premium per MW; paid for in schedule and certainty.

Prefabricated modular construction requires the structure and the principal interfaces to be fixed earlier than conventional construction does, but it does not rule out subsequent adjustment. On a case supplied by the project party, the Johor programme in Malaysia accommodated a request from Alibaba to adjust the internal functional layout. Whether a given change is feasible depends on whether it affects the load-bearing system, equipment already integrated, certification, or transport conditions; the effect on production schedule and cost has to be assessed in each case.

There is a second, less obvious difference. In a steel-frame building the critical path runs through the structure: nothing else can start until the frame is up and clad. In a prefabricated modular building the critical path runs through **the factory line and the foundations, in parallel**, and the two converge at the crane. That is the whole of the schedule advantage, and it is also the whole of the risk: a module that arrives before its plinth is ready, or a plinth built to construction rather than manufacturing tolerance, destroys the benefit immediately.

4 From factory line to live hall: the delivery sequence

The plates that follow document the sequence in a working prefabricated-module data centre programme, from the bare chassis on a factory line to a liquid-cooled hall in service. They are reproduced here because the argument of this paper — that integration has moved from the site to the factory, and that this is what makes stacking tractable — is easier to see than to assert. Read them as five stages: **manufacture, integration, assembly, plant, operation**.

STAGE 1 · MANUFACTURE — THE CHASSIS ON THE LINE



PLATE 1 The module begins as structure, not as a box. A welded steel chassis — base grillage, corner posts, roof frame — on a powered transfer pallet at station 28 of a production line, tare weight marked at 1,900 kg. The floor deck is already laid and drilled to the equipment setting-out. Everything that follows is added to this frame while it advances down the line, in a covered, lit, climate-controlled environment with overhead lifting on every station. This is the single most important image in the sequence: the object on the pallet is simultaneously the building's primary structure, its transport frame and its lifting rig.

STAGE 2 · INTEGRATION — SERVICES INSTALLED AND TERMINATED IN THE MODULE

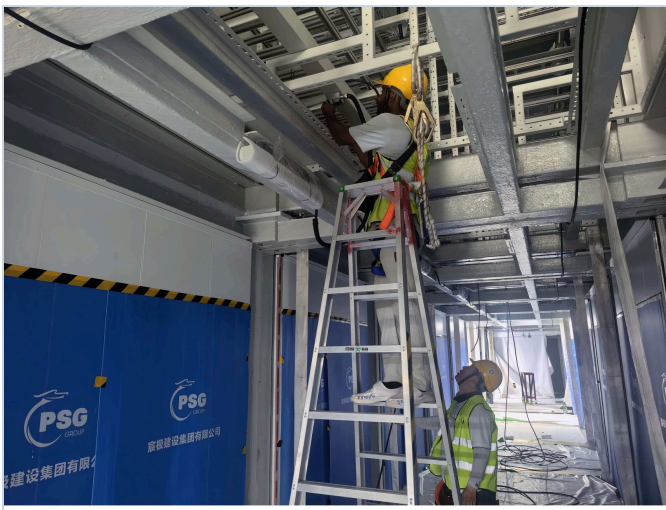


PLATE 2 Mechanical and electrical fit-out inside the module. Overhead containment, cable ladder, ductwork and hangers being installed at ceiling level within the module volume. Work at this stage is done from a stepladder in a protected space, to a repeated detail, rather than from a mobile platform in the open.

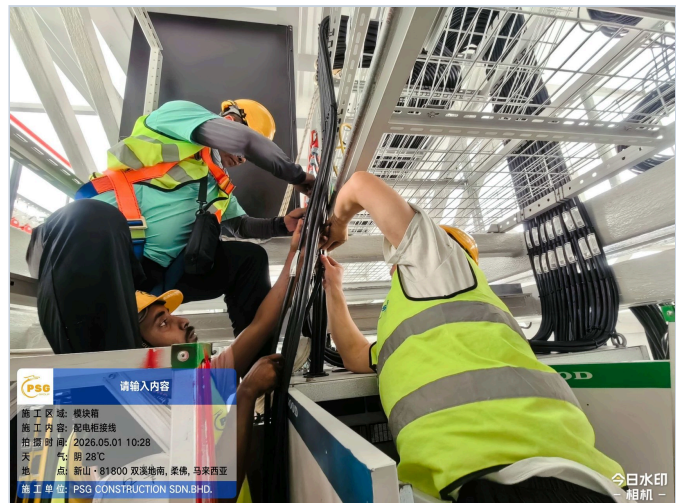


PLATE 3 Switchboard termination inside the module box. Power cabling dressed into distribution boards, with tray, busway and pre-formed cable sets overhead. The stamp records the work as 模块箱 (module box) / 配电柜接线 (switchboard wiring), Johor Bahru, Malaysia, executed by PSG Construction Sdn. Bhd.

**PLATE 4**

The pre-piped coolant gallery — the clearest single argument for factory integration. Stainless primary headers on a fabricated support frame, actuated control valves on every branch, flexible 30-bar coolant hoses to the rack-side manifolds, and segregated power and control containment below. In a site-built hall every one of these joints would be made outdoors by a different trade at a different time; here they are made on a jig, pressure-tested as an assembly, and delivered proven. A closed liquid loop is only as reliable as its worst field joint, which is why moving those joints indoors is not a convenience but a reliability strategy.

STAGE 3 • ASSEMBLY — SETTING THE BUILDING



PLATE 5

A data hall being assembled from prefabricated building modules. At the left, modules already set and joined, with the roof and plant deck being completed over them; in the centre, a crawler crane working the setting face; at the right and rear, several hundred completed modules in laydown awaiting their slot. This is what distinguishes the prefabricated building module from every other form of modular delivery: the units are not parked next to each other as separate machines, they are **knitted into one continuous building**. The laydown yard is, in effect, the project's inventory of finished floor area — and the reason the programme is limited by crane cycles rather than by trade availability.

STAGE 4 • PLANT — THE DECK ABOVE THE HALL



PLATE 6 **Completed envelope with the rooftop plant deck under installation.** The modules are set on plinths that lift the floor clear of the ground and form a continuous service undercroft beneath the hall; above, a steel deck carries cooling plant and the primary pipe racks. Power and coolant therefore arrive from below and heat leaves from above, over the shortest possible vertical path — the same logic that, taken one step further, produces the stacked hall discussed in section 9.

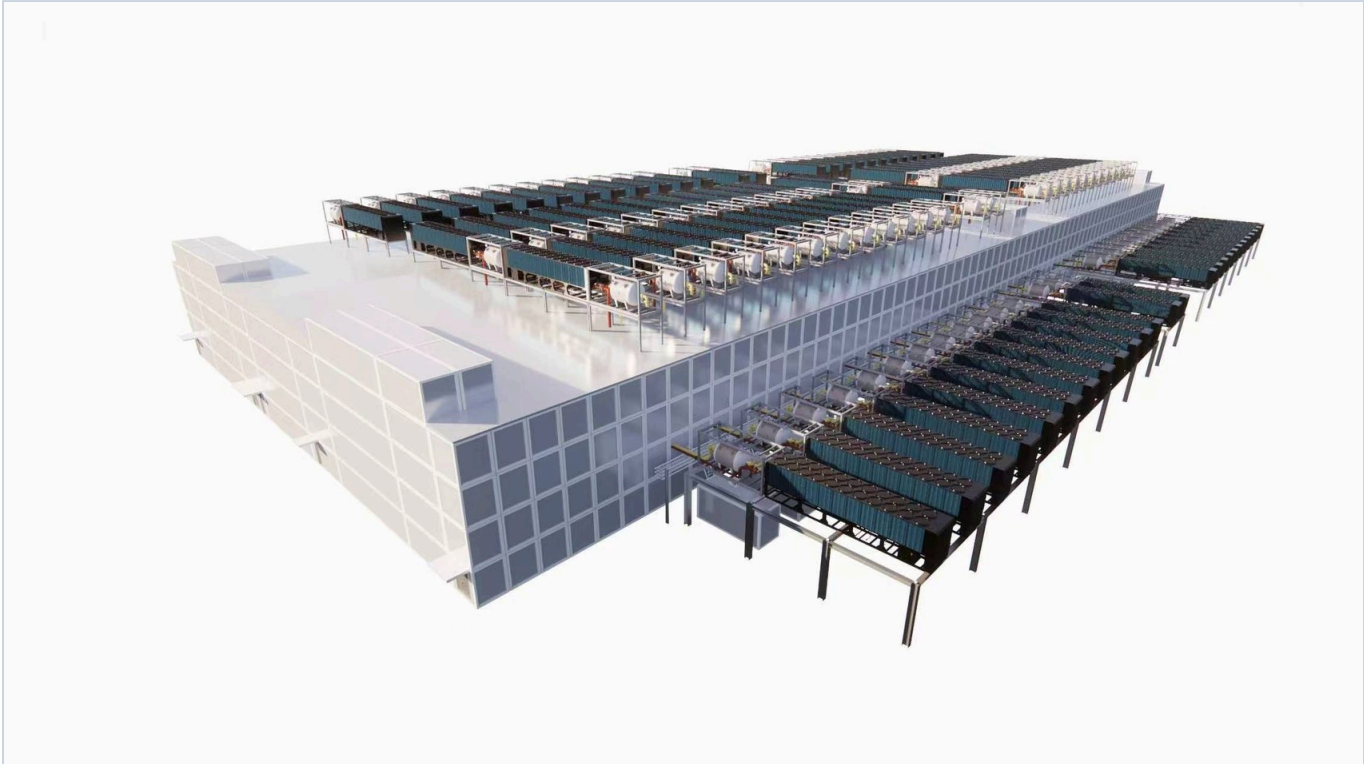


PLATE 7 **The configuration in general arrangement.** A modular hall with its IT volume enclosed at ground level and the full cooling plant carried on an elevated steel deck directly above and alongside it, arranged in repeating bays that mirror the module grid below. Vertical zoning of this kind — plant over IT, services dropping rather than running — is the intermediate step between a single-storey modular hall and a genuinely two-level stacked one.

STAGE 5 · OPERATION — THE LIQUID-COOLED HALL IN SERVICE

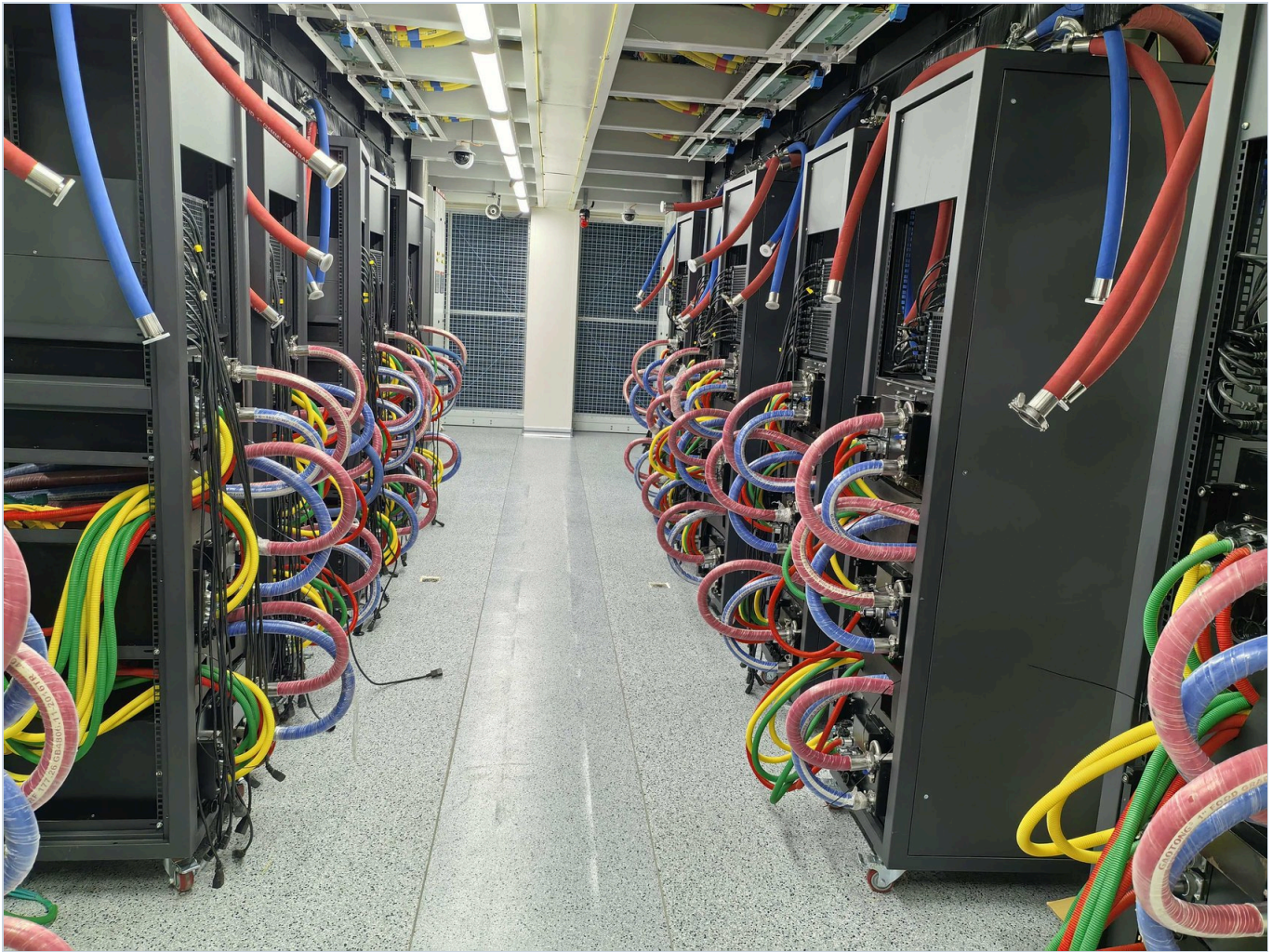


PLATE 8 A direct-to-chip liquid-cooled hall, live. Colour-coded supply and return hoses run from the rack-side manifolds to each cabinet, with overhead containment carrying power and network separately above the cold aisle. This is the load the building exists to serve: at 100–200 kW per cabinet the coolant distribution is no longer a service routed through the building — it is part of the building, and it is the reason the whole assembly is manufactured rather than constructed.

WHAT THE SEQUENCE SHOWS

Between Plate 1 and Plate 5, the proportion of the finished building produced under factory conditions is close to complete: structure, envelope, containment, switchboards, coolant headers and controls all arrive as one tested object. Site work reduces to four operations — **foundations and plinths** set to manufacturing tolerance, **craneage and setting**, **connection** of the four service families that cross the module boundary, and **integrated testing**, which becomes a verification exercise rather than a commissioning one because each unit was already proven.

The commercial consequence follows directly. Capacity arrives in module-sized quanta of 1–2 MW rather than in single 20–50 MW building steps, which lets construction be financed against a leasing curve instead of ahead of one.

5 The container-scale case, and why it is different

Container-scale MDC is the most constrained variant and, for that reason, the most instructive contrast. Its logic runs as follows.

The envelope is set jointly by sea and road transport conditions. A container-scale module has to satisfy the ISO box format, ship loading, port lifting and the final road leg at once, and its internal layout, aisles and service clearances are constrained accordingly. The 40 ft format is convenient precisely because it connects sea and land carriage, but adopting it

does not mean every leg of the journey is permit-free: each movement still has to be verified against the route, the vehicle combination and the total loaded weight.

The module is a sealed, pre-integrated system. Inside the envelope the manufacturer installs racks or rack positions, busway or distribution, UPS or battery interface, cooling distribution — CDUs, manifolds, direct-to-chip coolant loops or immersion tanks — containment, fire detection and suppression, monitoring and controls. The unit is powered, loaded with heaters simulating IT load, run through a factory acceptance test, and shipped as a tested system rather than a kit of parts.

Capacity arrives in quanta. Deployment proceeds in 0.5–2 MW increments matched to tenant demand, which is why container-scale modular has become the standard vehicle for sovereign and merchant AI capacity in the Middle East and Southeast Asia.

The difference from the prefabricated building module is architectural, and it is decisive at scale. A container is a **machine placed on a site**: it carries only itself, its walls are not a building line, and a yard of them does not become a hall. A prefabricated building module is a **piece of a building**: it is designed for module-to-module joints, for continuous fire and acoustic lines, for a shared roof and plant deck, and for superimposed load from a module above. That is why the container is the right answer for 2 MW at the edge and the wrong answer for 80 MW on a campus — and why every gigawatt-class programme in this paper's section 8 uses building modules, prefabricated plantrooms, or both.

6 Technological feasibility: what has been proven

Three questions determine whether prefabricated modular delivery is technically viable for AI workloads rather than merely for traditional colocation: can it carry the thermal load, can it satisfy structural and life-safety codes, and can it be certified repeatably.

Thermal

This is settled. Direct-to-chip cold plates, rear-door heat exchangers and single-phase immersion all now ship as sealed factory-integrated units capable of carrying current 700 W accelerators with roadmap headroom toward 1,000 W devices. Vendor claims of liquid-cooling-enabled PUE approaching 1.15 are credible in principle, though they should be read as design-point rather than annualised figures. The significant point is architectural, and Plates 4 and 8 make it: because the coolant loop is assembled and pressure-tested on a jig in a factory, prefabricated construction is arguably *better* suited to liquid cooling than site-built construction, where thousands of field-made joints in a closed loop represent a serious leak-risk surface.

Structural and seismic

Here feasibility is jurisdiction-dependent, and Japan is the strictest case in the world. Japanese data centre practice targets seismic performance at **importance factor 1.5**. It is worth being precise about the status of that figure, because it is widely misdescribed: **重要度係数 does not appear in the Building Standards Act**, and a privately owned data centre can lawfully obtain building confirmation at 1.0. The 1.5 / 1.25 / 1.0 ladder is the III · II · I classification of MLIT's 官庁施設の総合耐震・対津波計画基準, a standard whose own scope is government facilities; it reaches data centres because **JDCC's facility standard adopts 耐震性能I類相当 — the 1.5 grade — as its Tier 4 criterion**, after which tenants and developers specify it contractually. Importance factor 1.5 is therefore a **de facto market standard enforced through tenant specification and tier certification, not a building-control requirement**. What is statutory is the verification route: building confirmation plus structural calculation conformity review (構造計算適合性判定, “tekihan”) under Article 6-3, carried out by the prefectural governor or a designated agency, without whose certificate the confirmation certificate cannot issue. Two Japanese products demonstrate that prefabricated building modules clear this bar:

- **Daiwa House Module DPDC**, announced 8 December 2025 and on sale nationwide from 5 January 2026, is a steel-framed seismic-resistant building module of approximately **200 m² and 6.6 m height**, delivering **1–2 MW per module** on

high-voltage supply, built to **importance factor 1.5**, with its **structural frame in conformance with (準拠) JDCC Tier 4** for robustness and availability, and delivery approximately **one year** from contract against a conventional programme of several years.³

- **NTT Facilities**, with Nippon Steel Engineering, is developing a modular building method for hyperscale data centres that targets a reduction of **up to approximately 50%** in the period from application and design through to completion, for facilities in the **tens to hundreds of MW** range, with practical implementation targeted **within FY2028**.⁴

Certification

The repeatability question is answered by type approval. Where a module type is certified once and manufactured repeatedly, the approval burden collapses from per-building to per-type. Japan provides this through type conformity approval (型式適合認定, Art. 68-10) followed by certification of the type-component manufacturer (Art. 68-11), which are two sequential approvals rather than one. Canada certifies factory-built construction through **CSA A277**, and a binational safety standard dedicated to prefabricated modular data centres — **ANSI/CAN/UL 2755**, from UL Standards & Engagement — entered development in February 2026; the current technical document is UL 2755:2025, an Outline of Investigation rather than a full standard. The United States combines state-level factory certification with local installation permitting, increasingly around **ICC/MBI 1200, 1205 and 1210** — ANSI-approved American National Standards published jointly by the International Code Council and the Modular Building Institute, adopted in seven states and incorporated into the 2027 I-Codes. The European Union works through CE marking under the Construction Products Regulation, where EN 50575 cable fire classification is a genuine barrier for non-European suppliers. China has gone furthest, with the **T/CIEP 0205—2026** prefabricated data centre technical specification, which covers siting, foundations, technical requirements and test methods, and applies expressly to single-storey and multi-storey stacked prefabricated data centres.

7 The EPC layer: who actually assembles the building

Discussion of modular data centres tends to concentrate on the module manufacturer, which is the wrong end of the problem. Delivering a prefabricated-module AI campus requires three distinct capabilities, and only the first is widely available.

1. **Module manufacture** — steel fabrication, envelope, production-line discipline. Widely available from the modular-construction and shipping-container industries.
2. **In-module MEP integration** — switchboards, busway, containment, coolant headers, controls, terminated to fixed interfaces and factory-acceptance-tested. Available from data centre MEP contractors, but rarely at production-line cadence.
3. **Site EPC and assembly** — foundations and plinths to manufacturing tolerance, transport and craneage planning, module setting, the module-to-module joints, the four boundary connections per module, the plant deck, and integrated testing of the assembled building. **This is the scarce capability**, because it is the only one that cannot be bought as a product.

The third capability is where programmes are won or lost. A module that is perfect in the factory is worth nothing if the plinth is 40 mm out, if the crane cycle is slower than the delivery convoy, or if the joint between two modules is detailed as a site improvisation. It is also the capability with the fewest credible providers at gigawatt scale, because it demands a contractor that can run a manufacturing-tolerance site, a heavy-lift programme and a data centre commissioning regime at the same time, in the jurisdiction where the building will stand.

The market for that capability is narrow enough to be listed. The table below reproduces the ranking of modular data centre site-assembly and mechanical-and-electrical construction contractors (模块化DC现场总装与机电施工、EPC) from the supplied industry briefing.

MODULAR DATA CENTRE SITE ASSEMBLY AND MEP CONSTRUCTION (EPC) — LEADING CONTRACTORS				
RANK	CONTRACTOR	HEADQUARTERS	CORE CAPABILITY	REPRESENTATIVE CREDITS
1	Comfort Systems USA	United States	Leading North American modular integrator; tied to hyperscale CSP demand	Backlog of US\$14.06 billion at 30 June 2026, up 73% year on year
2	PSG Construction	Japan / Malaysia	Containerised and modular DC mechanical-and-electrical main contracting	Johor Bahru, Malaysia — 42 MW (in delivery); Osaka, Japan — approx. 2.5 MW (2025); United States
3	Multiplex	Australia	Large-scale DC main contracting; prefabricated plantrooms and precast lift cores	NEXTDC S4 Sydney — multi-storey, completion mid-2027 ²
4	HOCHTIEF Polska	Poland	DC main contracting; approximately 60% of components factory-prefabricated	Data4 Jawczyce, Poland
5	GS E&C	Korea	EPC party to the PMDC alliance; large-scale DC construction experience	LG U+ PMDC alliance reference model; constructability review

Two entries corroborate material used elsewhere in this paper. Multiplex at rank 3 is the contractor on NEXTDC S4 Sydney, cited in sections 1 and 8; HOCHTIEF Polska at rank 4 is the contractor behind the Data4 Jawczyce prefabrication share quoted in section 8. That the same firms appear independently in both the project record and the capability ranking is a modest but real check on the ranking's plausibility.

GRID APEX'S EPC CONTRACTOR

PSG Construction is Grid Apex's EPC contractor for the campus — engaged for capability 3 above, site assembly and construction, together with the in-module mechanical-and-electrical integration of capability 2. The briefing ranks the firm **second globally** in this category, and it is headquartered across **Japan and Malaysia**, which matters more than the ranking does: the Japanese presence is what allows a module manufactured and integrated offshore to be received, set and certified on a Japanese site under Japanese supervision, and the Osaka project establishes that the firm has already built modular data centre capacity inside the Japanese regulatory regime.

Plates 2, 3, 5 and 6 in section 4 are from PSG Construction's Johor programme. Plate 3 carries the contractor's own site record — work area 模块箱 (module box), scope 配电柜接线 (switchboard wiring), Johor Bahru, executed by **PSG Construction Sdn. Bhd.** — and Plate 5 shows that programme assembling a continuous data hall from several hundred prefabricated building modules, with a laydown yard of completed units feeding a single setting face. The capability being contracted for Project MIE is therefore one that has been photographed in execution, not one that has been described in a capability statement.

For Project MIE this matters in a specific way. The Japanese constraint set — the importance-factor-1.5 market standard, structural conformity review, a construction labour market that cannot staff a conventional programme, and an architect-of-record regime that supervises on site — rewards a delivery model in which the maximum possible share of the building is manufactured and tested offshore under accredited conditions, and the minimum possible share is built in the field. That model only works if the EPC contractor at the receiving end can hold manufacturing tolerance on a Japanese site under Japanese supervision. The contractor selection is therefore not a procurement decision downstream of the design; it is part of the design — which is why Grid Apex has made it, and made it early.

8 Example projects across nine markets

REPRESENTATIVE MODULAR AI DATA CENTRE PROJECTS		
MARKET	PROJECT	WHAT IT DEMONSTRATES
United States	Snyder AI Campus, Texas — Energy Vault, using Crusoe Spark modules	Powered-land model: generation, storage and modular data centre co-located. Ground broken July 2026; phase 1 of 8 MW, campus designed for up to 500 MW, with Crusoe Cloud as anchor tenant.
Canada	DMG — Christina Lake, British Columbia	Prefabricated secure data centre units delivered under the new CSA prefabricated-modular framework.
Europe	Data4 — Jawczyce, Poland; CADOLTO — NorthC	Roughly 60% factory prefabrication of components on a hyperscale site; and two-storey modular data halls using stacked 750 kW building modules.
Japan	Daiwa House Module DPDC; NTT Facilities modular building method (with Nippon Steel Engineering)	Prefabricated building modules clearing the world's strictest seismic regime — at one to two storeys .
Korea	LG U+ PMDC alliance (with GS E&C, Zay C&A, Gansam and D&O CM); kt cloud 1 GW target	Consortium standardisation, MOU September 2026: a 100 MW-class reference model covering both new build and retrofit of existing buildings; alongside kt cloud's 1 GW AI data centre capacity target for 2031.
Southeast Asia	Johor, Malaysia — prefabricated building-module data halls (PSG Construction Sdn. Bhd., plates 2, 3, 5, 6); MRCB + TECO, Malaysia; CIMC prefabricated data hall deliveries into Indonesia	Continuous data halls assembled from several hundred building modules; highest regional acceptance of stacked module architecture, driven by heat, humidity and land cost.
Middle East	Khazna Ajman, UAE; Mindstream Al-Risha, Jordan	Modular scalable architecture — 100 MW across 20 data halls of 5 MW each, Uptime Tier III design certified; and a 400 MW sovereign AI platform using single-phase immersion (dielectric) liquid cooling.
Australia	NEXTDC S4, Western Sydney (Multiplex)	350 MW across two four-storey buildings , a 30,743 m ² facility, with prefabricated plantrooms, switch rooms and services reticulation; mid-2027. ²
China	Huawei 3D Data Centre; Alibaba CUBE 5.0, Ulanqab; CIMC (>1,000 MW cumulative, per CIMC)	Four-layer vertical stacking in production; and programme compression from 6–12 months to around 100 days.

9 The stacked data hall — and the two-level case

Stacking is where prefabricated modular construction stops being a procurement strategy and becomes an architectural one. The motivation is straightforward: on a land-constrained or power-dense site, megawatts per hectare is the governing metric, and the only way to raise it without raising rack density further is to build upward. It is also the point at which the container and the building module part company for good — only a module engineered for superimposed load can carry another module.

The proof point

The most significant demonstration to date is **Huawei's 3D Data Centre**, unveiled on 15 September 2026 at the AIDC Industry Development Conference in Wuhu, Anhui. It organises the facility into **four vertically stacked functional layers** — cooling, IT equipment, power supply and backup power — rather than distributing those functions horizontally across a site. It is architected to support **Ascend supernode clusters of 100,000 cards and above** (十万卡以上), which Huawei presents as the coming baseline configuration rather than as one building's nameplate. The justification offered is the trajectory of cabinet

power, which the industry has seen rise from a few kilowatts in the air-cooled era to **100–200 kW** and which continues to climb; the vertical arrangement is presented as the answer to three simultaneous constraints: space, power delivery and thermal management.⁵

Plates 6 and 7 show the same logic one step earlier in its development: the IT volume at ground level, the entire cooling plant on a deck directly above it, and services dropping vertically into the hall rather than running horizontally across it. Going from that arrangement to a true two-level stack means replacing the plant deck with a second occupied module level — a structural change, not a services one.

The proof in assembly

Stacked volumetric modules are not a drawing exercise. The two plates below record a multi-storey prefabricated modular building being assembled from the same class of unit described in section 2 — steel-chassis volumetric modules set one on another and knitted into a continuous structure.



PLATE 9 **Five levels of stacked volumetric modules.** The module grid reads directly in the elevation: each bay is one factory-built unit, with its own chassis posts and edge beams, set on the unit below and spliced at the corners. The pale blue members are the modules' own structural frames; the panelised infill between them is envelope, not structure. Every module carries its share of the superimposed load from the levels above — which is precisely the load case that separates a building module from a shipping container, and the case that must be verified before any two-level data hall can be approved.



PLATE 10 **Stacking in progress, over a foundation grid built to manufacturing tolerance.** In the foreground, the prepared substructure: a regular grid of reinforced strip footings and pile caps with starter bars, set out to the module grid so that every unit lands on a prepared bearing. Behind it, modules already set two and three high, with a crawler crane and a tower crane working the setting face and the next units laid out beside them. This single frame contains the whole discipline of the method — the substructure is surveyed and cast to millimetres because the superstructure arrives finished and cannot be adjusted to fit.

WHY VERTICAL ZONING WORKS

- **Gravity is free.** Placing cooling plant above the IT floor lets coolant return by gravity and shortens the hydraulic path — the dominant parasitic load in a liquid-cooled facility.
- **Services travel vertically, not horizontally.** A 200 kW rack needs power and coolant delivered over the shortest possible distance. Stacking converts long horizontal runs into short vertical drops, cutting conductor mass, pumping energy and pressure drop.
- **Fault propagation is bounded by the floor slab.** A structural deck is already a fire and water barrier. Vertical zoning gives compartmentation for free — a leak in a cooling layer does not reach IT on a different level.
- **Maintenance separates from operations.** Plant on its own level can be serviced without entering the data hall, which is what concurrent maintainability requires in practice.

Why two levels specifically

Two levels captures most of the benefit at a fraction of the regulatory and structural cost, for four reasons.

The structure stays conventional. A two-level assembly of prefabricated building modules can be built as modules-on-modules or as modules set into a simple frame, with load paths a structural engineer can verify by inspection. Beyond two or

three levels, the lower modules become primary structure carrying substantial superimposed load, the frame becomes a moment-resisting system in its own right, and the design leaves the realm of repeatable type approval.

Seismic demand scales faster than height. Under a seismic regime, mass at height attracts lateral force, and the overturning moment grows with the square of height. At two levels the centre of mass stays low and foundations remain economical. This is the binding constraint in Japan and the reason its market has settled where it has.

Fire strategy remains tractable. Two levels permits direct external escape from both, straightforward fire-brigade access and simple smoke control. Higher stacks require engineered smoke management and protected escape routes — deliverable, but no longer a repeatable factory product.

Density roughly doubles without changing the module. The same certified module, the same transport envelope, the same factory line, the same setting cycle — twice the megawatts per hectare. The efficiency of prefabricated delivery is preserved exactly.

What the markets accept

STRUCTURAL ACCEPTANCE OF STACKED MODULAR DATA HALLS BY MARKET			
MARKET	STACKING ACCEPTED	GOVERNING CONSTRAINT	EVIDENCE
China	Multi-level, no published limit	T/CIEP 0205—2026 applies expressly to single-storey and multi-storey stacked prefabricated data centres; no numeric ceiling is published	Huawei 3D DC — four layers in production
Korea	High	Consortium standard models rather than statute	100 MW-class PMDC reference model
Southeast Asia	High	Land cost and climate favour density	Stacked module architecture common in Malaysia and Indonesia
Middle East	High	Speed and sovereign AI strategy	Khazna Ajman — modular scalable architecture
United States	High	IBC plus state factory certification	Powered-land campuses scaling 8 MW → 500 MW (Snyder, Texas)
Australia	Moderate–high	Conventional frame with prefabricated MEP	NEXTDC S4 — two four-storey buildings ²
Europe	Moderate	CPR and EN 50575 fire classification	CADOLTO NorthC — two-storey modular halls, stacked 750 kW modules
Canada	Moderate	CSA A277 plus provincial approval	Prefabricated units, typically 1–2 MW
Japan	Limited — 1 to 2 levels	Structural conformity review, plus an importance-factor-1.5 market standard; the most demanding regime globally	NTT Facilities explicitly targets 1–2 storeys ⁴

10 The Japanese case

Japan is simultaneously the hardest market in which to stack and one of the most compelling in which to build with prefabricated building modules — and the two facts share a cause.

The drivers are acute. Construction labour is severely constrained. Grid connection in the Tokyo core is reported to carry waiting times of five to ten years, which pushes projects to regional sites where local trade capacity is thinner still. Against

that, a one-year delivery programme is not a convenience; it is the difference between a viable project and an unviable one. Every hour of work moved from a Japanese site into an accredited factory is an hour removed from the binding constraint.

The specification and review constraints are equally acute. Importance factor 1.5 is not a legal requirement, as section 6 sets out, but it is not optional either: no hyperscale tenant will accept a Japanese data hall below it, and JDCC Tier 4 cannot be claimed without it. Structural conformity review then adds a statutory step that runs to a fixed clock and that rewards conventional, verifiable structural forms while penalising novelty. A five-level modular stack in Japan would face a structural review with little precedent, a more complex fire strategy, and an approval timeline that would consume the schedule advantage that justified modular construction in the first place.

The Japanese industry's answer is consistent across its two most credible products, and it is instructive: **keep the building low and solve the seismic problem with isolation rather than with mass**. NTT Facilities' approach is explicit — rather than base-isolating the entire structure, it isolates **the data hall floor only** while applying vibration control to the main equipment, maintaining base-isolated-equivalent business continuity performance at one to two storeys.⁴ Daiwa House takes the complementary route: a single-level 200 m² steel building module engineered to importance factor 1.5, with its structural frame in conformance with JDCC Tier 4, avoiding stacked structural evaluation altogether.³

There is one further Japanese consideration that bears directly on the module supply chain. Structural steel must be a designated building material conforming to a JIS named in Notification 1446 of 2000 under Article 37, item (i) of the Building Standards Act, or else carry individual ministerial approval under item (ii) — the route that applies where fabrication sits outside a certified JIS mill. SS (JIS G 3101) and SM (G 3106) grades remain lawful; **SN grades (JIS G 3136, typically SN400B and SN490B) are the Japanese design convention and effectively mandatory in practice for seismic-resisting and welded members**, because only SN specifies yield-ratio caps, impact values and weldability control. A conformity-review examiner will expect them there; and repeatable factory manufacture of a certified module type runs through **two sequential approvals** — type conformity approval of the design (型式適合認定, Art. 68-10), then certification of the manufacturer (Art. 68-11, or Art. 68-12 where the factory is overseas, as it will be here). Manufacturer certification carries five-year validity and requires factory work to be at least two-thirds of the combined production-and-installation process. **No standard processing period is published** for either approval by MLIT or any designated certification body, and the designated bodies direct applicants to pre-application consultation instead, so the programme must be fixed by consultation rather than assumed. Both are answerable, but both must be started early and neither can be retrofitted to a module already in production. This is the practical form the “freeze the design early” discipline takes in Japan.

THE SYNTHESIS FOR A JAPANESE AI CAMPUS

The evidence points to a specific and defensible architecture for a large Japanese AI data centre: **low-rise prefabricated building modules — one to two levels — combined with a hybrid isolation strategy** in which the structure is highly seismic-resistant, primary plant is vibration-controlled, and the data hall floor is base-isolated; manufactured under type conformity approval and assembled by an EPC contractor able to hold manufacturing tolerance on a Japanese site — for Grid Apex, PSG Construction.

This configuration captures the density and service-routing benefits of vertical zoning, stays within the structural acceptance envelope that Japanese reviewers have demonstrably approved, preserves type-approval repeatability, and keeps the schedule advantage intact. Attempting four or five levels in Japan would trade a proven one-year programme for an unproven multi-year approval — which is precisely the trade modular construction exists to avoid.

11 Grid Apex's adopted approach

Grid Apex is building its multi-storey AI data centre campus using prefabricated building modules. The campus is designed as two-level stacked data halls assembled from factory-manufactured, factory-tested structural modules, set on permanent foundations, joined into continuous buildings, and served by a rooftop plant deck — the configuration argued for throughout this paper and illustrated in plates 1 to 10. The modules are manufactured and integrated offshore under accredited

conditions, verified against Japanese structural, electrical and fire requirements before shipment, and assembled on the Japanese site by **PSG Construction**, Grid Apex's appointed EPC contractor for the campus, under the supervision of the Japanese architect of record.

Two questions follow from that decision, and both are asked by every investor and every tenant. Is the result a permanent building or a temporary installation? And can a factory-built module house the current and next generations of AI hardware? The answers are, respectively, a matter of law and structure, and a matter of how the capacity envelope is specified.

Why the building is permanent

“Modular” is often heard as “temporary”, because the word is associated with site cabins and container yards. A prefabricated building module campus is a different object in every respect that matters — legally, structurally and financially. The distinguishing test is simple: **the module is not a machine standing on the ground, it is part of a building fixed permanently to it.**

PERMANENCE: TEMPORARY MODULAR INSTALLATION AGAINST THE GRID APEX APPROACH		
TEST	TEMPORARY / CONTAINER INSTALLATION	PREFABRICATED BUILDING MODULES — GRID APEX
Legal status	Often consented as equipment, a structure (工作物) or a temporary building with a limited consent period.	A building (建築物) under the Building Standards Act, with full building confirmation, structural calculation conformity review and completion inspection certificate.
Foundations	Set on pads, blocks or a slab; removable without demolition.	Piled or reinforced strip foundations cast to the module grid, with cast-in starter bars and anchor assemblies (Plate 10). Removal is demolition, not relocation.
Structure	Each unit carries only itself; units are placed adjacent, not connected.	Module frames are spliced into one continuous lateral load-resisting system carrying superimposed load from the level above (Plate 9).
Seismic basis	Equipment-grade anchorage.	Structural verification through building confirmation and 適判, evaluated for the stacked configuration, and designed to the importance factor 1.5 market standard that JDCC Tier 4 requires.
Envelope and fire	Weather protection; fire strategy unit by unit.	Fire-rated, insulated building envelope with continuous compartmentation lines across module joints and a whole-building fire strategy.
Design life	Typically 10–15 years.	Specified to conventional building design life, on the same steel and protective-coating basis as a site-built steel-frame data centre.
Asset treatment	Plant and equipment; short depreciation; financed as equipment.	Real property: registered, depreciated on a building schedule, insurable and financeable as a building, and mortgageable as security.

The point that resolves the question for most readers is the inspection trail. A Grid Apex module does not enter Japan as freight that is then parked; it enters as a certified building component. It is manufactured under type conformity approval (型式適合認定) with accredited type-component manufacture, from JIS-conforming structural steel, against a design that has already passed structural conformity review. On site it is set on permanent foundations, spliced, sealed, connected and then subjected to **the same completion inspection as any other building**, issuing the same certificate. The regulatory file at the end of the process is indistinguishable from that of a conventionally constructed data centre, because the regime is the same regime. What differs is where the work was done, not what standard it was done to.

Three consequences follow, and they are the ones that matter commercially. The building can be **mortgaged and insured as real property**, which is what makes a ten-year lease financeable. It can be **depreciated over a building life** rather than an

equipment life, which is what makes the economics work. And it **cannot be repossessed or relocated**, which is precisely the assurance a tenant committing to a decade of capacity requires.

Why it houses the latest AI equipment

The second question is usually asked as a doubt about factory construction. It is better understood as a question about **which envelope is fixed and which is not**. A data centre does not need to anticipate the specific accelerator it will host; it needs a shell whose four capacity envelopes — floor loading, power path, coolant path and heat rejection — are specified above the hardware roadmap rather than to the hardware of the day. Prefabricated construction is advantaged here, not disadvantaged, because those four envelopes are engineered into a repeated, verified module instead of being discovered during a site fit-out.

THE FOUR CAPACITY ENVELOPES, AND HOW THE MODULE PROVIDES EACH		
ENVELOPE	WHAT THE LATEST AI HARDWARE DEMANDS	HOW THE PREFABRICATED BUILDING MODULE ANSWERS IT
Floor loading	A fully populated liquid-cooled AI rack is a concentrated load of roughly 1.5–2.0 tonnes on a footprint under 1 m², in continuous rows.	The module deck is structural steel designed for the rack layout from the outset, with the load path from rack foot to beam to column to foundation drawn before manufacture. There is no raised-floor void to bridge and no post-hoc strengthening.
Power path	100–200 kW per cabinet today, with a roadmap pointing at 600 kW; higher-current busway, larger conductors, more switchgear per square metre.	Power terminates at fixed interfaces on the module face. Uprating a hall means changing distribution inside modules — a factory-repeatable operation — not rebuilding risers through an occupied building.
Coolant path	Direct-to-chip liquid cooling at 700–1,000 W per device, with rack-side manifolds, secondary loops and CDU capacity, and leak tolerance approaching zero.	Primary headers, valved branches, CDU positions and manifold connections are installed on a jig and pressure-tested as an assembly before shipment (Plate 4). The hall is liquid-ready on day one rather than retrofitted.
Heat rejection	Every watt delivered must leave; plant capacity, not IT space, is the true constraint on density.	The plant deck sits directly above the hall on its own structure (Plates 6 and 7), sized to the module grid below and extendable bay by bay as density rises.

The governing insight is one of timescales. **IT hardware turns over every two to four years; the building lasts decades.** That mismatch is not created by modular construction — it is the central fact of every data centre ever built. What modular construction changes is the cost of living with it. Because the IT boundary is a defined set of interfaces at the module face rather than a web of site-made connections, a hall can be taken out of service, re-fitted to a new rack generation, and returned to service **module by module, without touching the structure, the envelope or the neighbouring halls**. The shell that receives Vera Rubin-class hardware in 2028 is the same shell that receives its successor, and the one after that.

THE GRID APEX POSITION

The campus is a **permanent Japanese building** — confirmed, conformity-reviewed, inspected and certified under the Building Standards Act, designed to the importance factor 1.5 that JDCC Tier 4 requires, on permanent foundations, financeable and insurable as real property — that happens to have been **manufactured rather than constructed**.

It is specified against the hardware roadmap rather than against today's hardware, on all four capacity envelopes, and it is designed so that the equipment inside can be replaced a generation at a time without the building being disturbed. It is delivered by an EPC contractor — PSG Construction — already ranked among the leaders in this exact scope and already building modular data centre capacity in Japan. Prefabrication is the delivery method. Permanence and capability are design requirements, and neither is traded away to obtain the schedule.

12 Conclusion

Prefabricated modular delivery of AI data centres is no longer a question of technical feasibility. The thermal case is closed: liquid cooling at 700–1,000 W per accelerator ships as a factory-integrated, pressure-tested assembly, and factory integration is a genuine advantage rather than a compromise. The structural case is closed in every major market, including the most demanding: the importance-factor-1.5 standard that Japanese tenants demand has been met by at least two commercially available building-module products. The certification case is closing fast, through type approval in Japan, CSA in Canada, state certification in the United States, CPR in Europe and T/CIEP in China.

What the plates in section 4 add to that is a reminder of where the residual risk actually sits. It is not in the module; the module is a manufactured product with a factory acceptance test behind it. It is in the **joint between the factory and the site** — plinth tolerance, transport and crane sequencing, module-to-module detailing, and the four boundary connections per unit. That is an EPC problem, and it is why the contractor capable of executing it is a design input rather than a procurement outcome.

What remains genuinely open is **how high**. Vertical stacking of AI data halls is proven in production at four levels in China, is being delivered at four storeys in Australia within a conventional frame, and is commercially available at two storeys in Europe. Japan sits at the conservative end of that distribution — not because stacking is technically impossible there, but because the seismic review regime makes each additional level disproportionately expensive in time.

For a large AI campus in Japan, the two-level stacked data hall built from prefabricated building modules is therefore the point of maximum advantage: roughly double the megawatts per hectare, shorter vertical service runs, compartmentation given by the structure, and maintenance separated from operations — all within a structural form that the Japanese approval system has already accepted. It is the configuration that takes what prefabricated construction is good at and stops exactly where the regulatory cost curve turns upward.

That is the configuration Grid Apex has adopted. The result is not a temporary installation and not a container yard: it is a permanent building, confirmed and inspected under the same Japanese regime as any site-built data centre, financeable and insurable as real property, and specified on all four capacity envelopes against the hardware roadmap rather than against the hardware of the day. Prefabrication is how it is delivered. What it delivers is a conventional promise — a durable building around equipment that will be replaced several times within its life — met several years sooner than conventional construction can meet it.

Sources and confidence. Every footnoted reference and every statistic, date, standard number and project credit in this paper was checked against primary sources on 22 September 2026. Claims fall into three tiers. **Verified against primary or named sources:** the market figures in the introduction (Fortune Business Insights, report 100504); Huawei's 3D Data Centre architecture and event; Daiwa House Module DPDC specification, announcement and on-sale dates; NTT Facilities' modular building method; NEXTDC S4 Sydney configuration; Comfort Systems USA backlog at 30 June 2026; Data4 Jawczyce and CADOLTO / NorthC; the Snyder, Christina Lake, Khazna Ajman, Mindstream, LG U+ PMDC, Alibaba CUBE 5.0 and MRCB / TECO projects; and the Japanese regulatory material in sections 2, 6 and 10 — the Vehicle Restriction Order limits, the status of importance factor 1.5, structural calculation conformity review, Articles 68-10 to 68-12, Article 37 and JIS G 3136, and the JDCC facility standard. **Documented in the photographic record** (plates 1–10, supplied by the client from live prefabricated-module programmes): the factory chassis and production line, in-module MEP and switchboard integration, the pre-piped coolant gallery, site assembly from several hundred building modules, the rooftop plant deck, a liquid-cooled hall in service, and multi-storey stacked volumetric module assembly over a module-grid substructure; the contractor identification in Plate 3 is taken from the site record stamped on the photograph itself. **Drawn from a supplied industry briefing and not independently verified:** the 20–30% per-MW cost premium, vendor PUE claims, regional policy detail, the contractor ranking positions in section 7 and PSG Construction's project capacities, CIMC's cumulative delivered capacity (a company statement), and, in section 3, the layout-change case at Johor reported by the project party. **Design intent rather than completed fact:** section 11 states Grid Apex's adopted approach, its appointment of PSG Construction as EPC contractor, and the design requirements set for its campus; the regulatory, structural and capacity characteristics described there are design targets to be evidenced through building confirmation, structural conformity review and completion inspection in the normal way, and are not claims about a building already certified.

References. 1 — Fortune Business Insights, *Modular Data Center Market*, 2025–2034. 2 — Multiplex, *Work commences on NEXTDC's 350MW Western Sydney Data Centre*. 3 — 大和ハウス工業, モジュール型データセンター商品「Module DPDC」販売開始, 8 December 2025. 4 — NTTファシリティーズ, ハイパースケイラー向けデータセンターにおけるモジュール型建築手法の開発に着手, 1 July 2026. 5 — Huawei 3D Data Centre, AIDC Industry Development Conference, Wuhu, 15 September 2026.